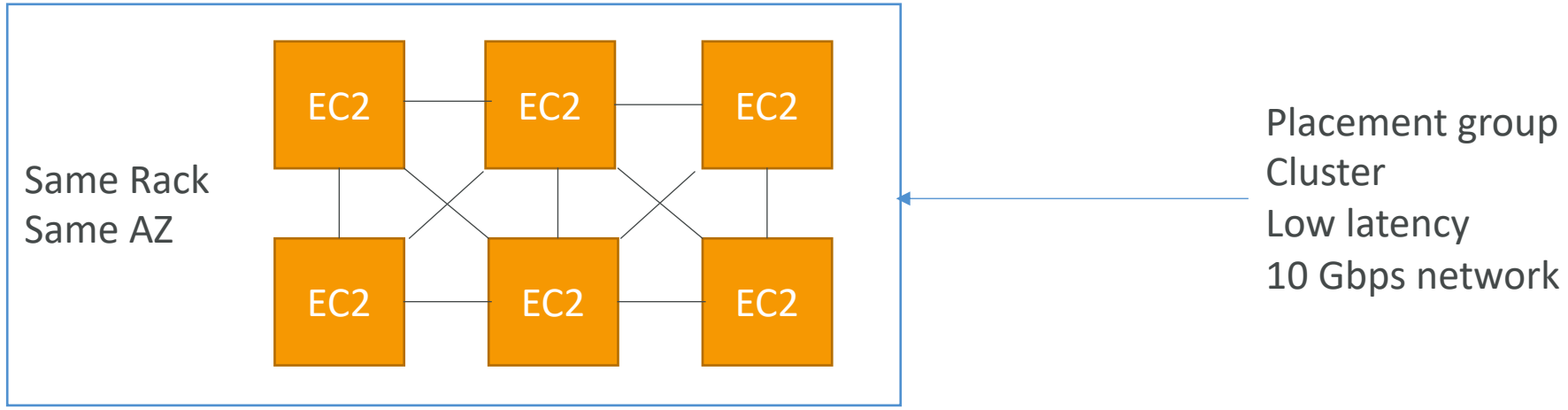
Amazon EC2 – Associate

Placement Groups

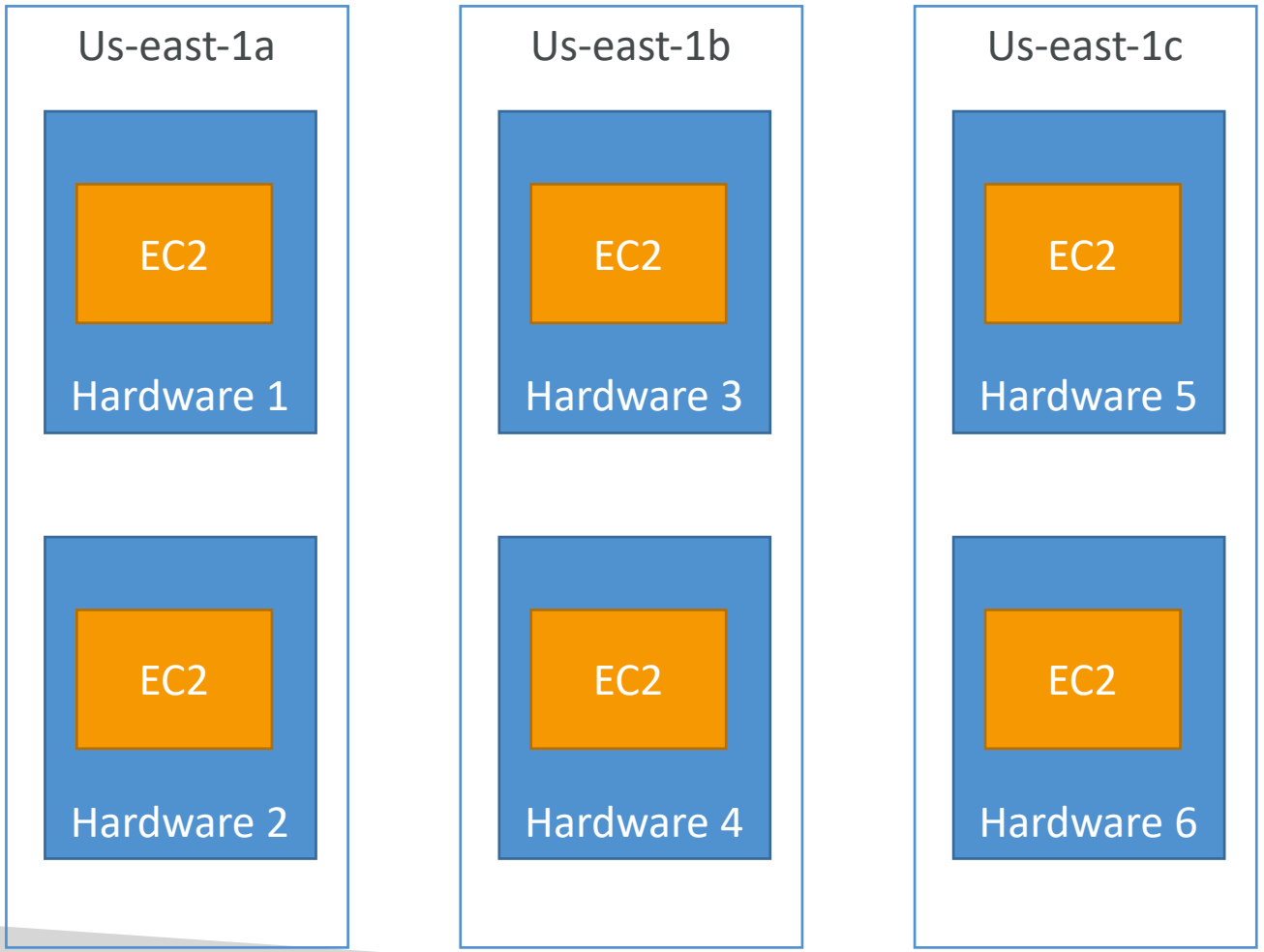
- Sometimes you want control over the EC2 Instance placement strategy
- That strategy can be defined using placement groups
- When you create a placement group, you specify one of the following strategies for the group:
  - *Cluster*—clusters instances into a low-latency group in a single Availability Zone
  - *Spread*—spreads instances across underlying hardware (max 7 instances per group per AZ)
  - *Partition*—spreads instances across many different partitions (which rely on different sets of racks) within an AZ. Scales to 100s of EC2 instances per group (Hadoop, Cassandra, Kafka)

Placement Groups
Cluster



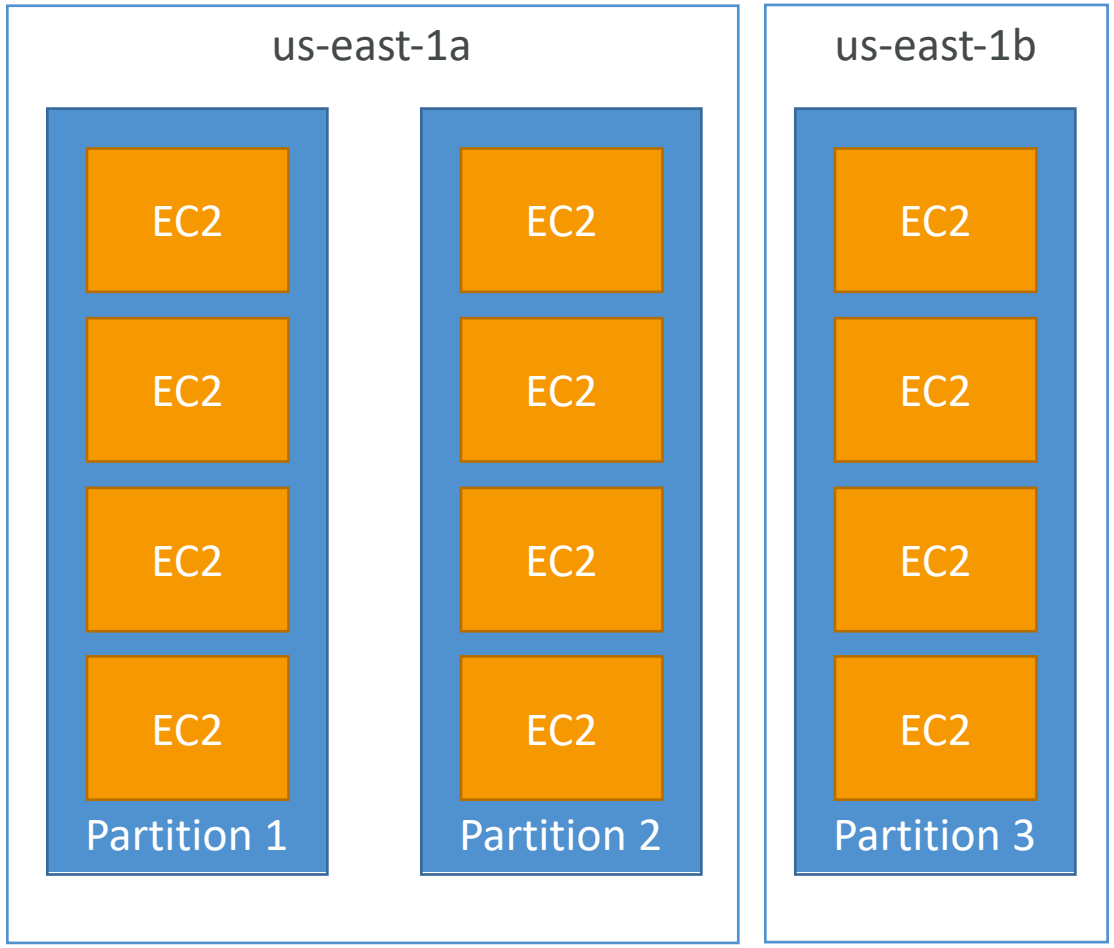
- Pros: Great network (10 Gbps bandwidth between instances with Enhanced Networking enabled - recommended)
- Cons: If the rack fails, all instances fails at the same time
- Use case:
  - Big Data job that needs to complete fast
  - Application that needs extremely low latency and high network throughput

Placement Groups Spread



- Pros:
  - Can span across Availability Zones (AZ)
  - Reduced risk is simultaneous failure
  - EC2 Instances are on different physical hardware
- Cons:
  - Limited to 7 instances per AZ per placement group
- Use case:
  - Application that needs to maximize high availability
  - Critical Applications where each instance must be isolated from failure from each other

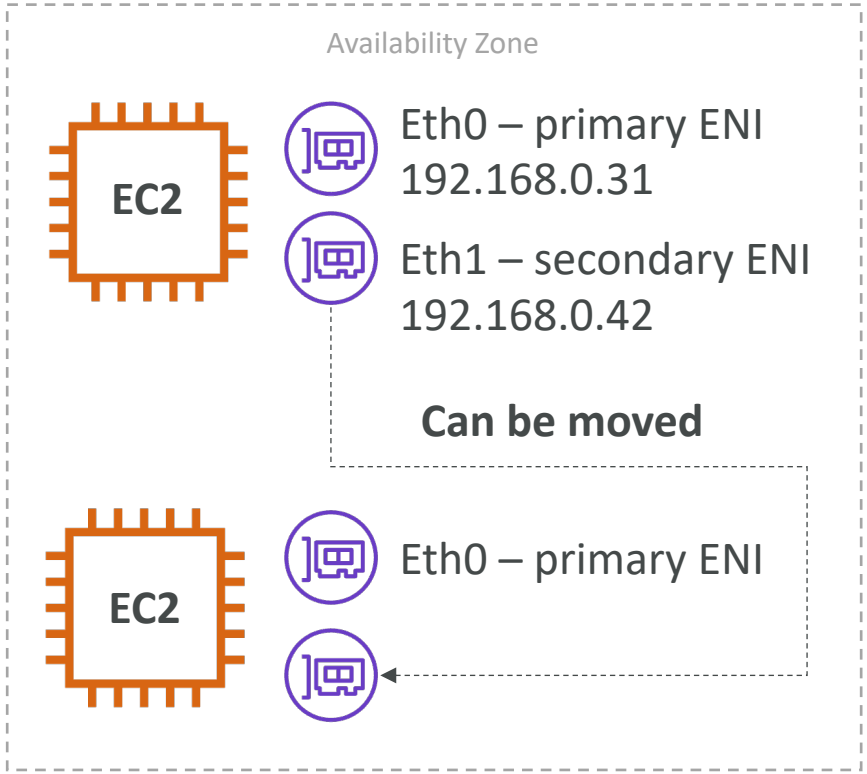
Placements Groups Partition



- Up to 7 partitions per AZ
- Can span across multiple AZs in the same region
- Up to 100s of EC2 instances
- The instances in a partition do not share racks with the instances in the other partitions
- A partition failure can affect many EC2 but won't affect other partitions
- EC2 instances get access to the partition information as metadata
- Use cases: HDFS, HBase, Cassandra, Kafka

Elastic Network Interfaces (ENI)

- Logical component in a VPC that represents a **virtual network card**
- The ENI can have the following attributes:
  - Primary private IPv4, one or more secondary IPv4
  - One Elastic IP (IPv4) per private IPv4
  - One Public IPv4
  - One or more security groups
  - A MAC address
- You can create ENI independently and attach them on the fly (move them) on EC2 instances for failover
- Bound to a specific availability zone (AZ)



EC2 Hibernate

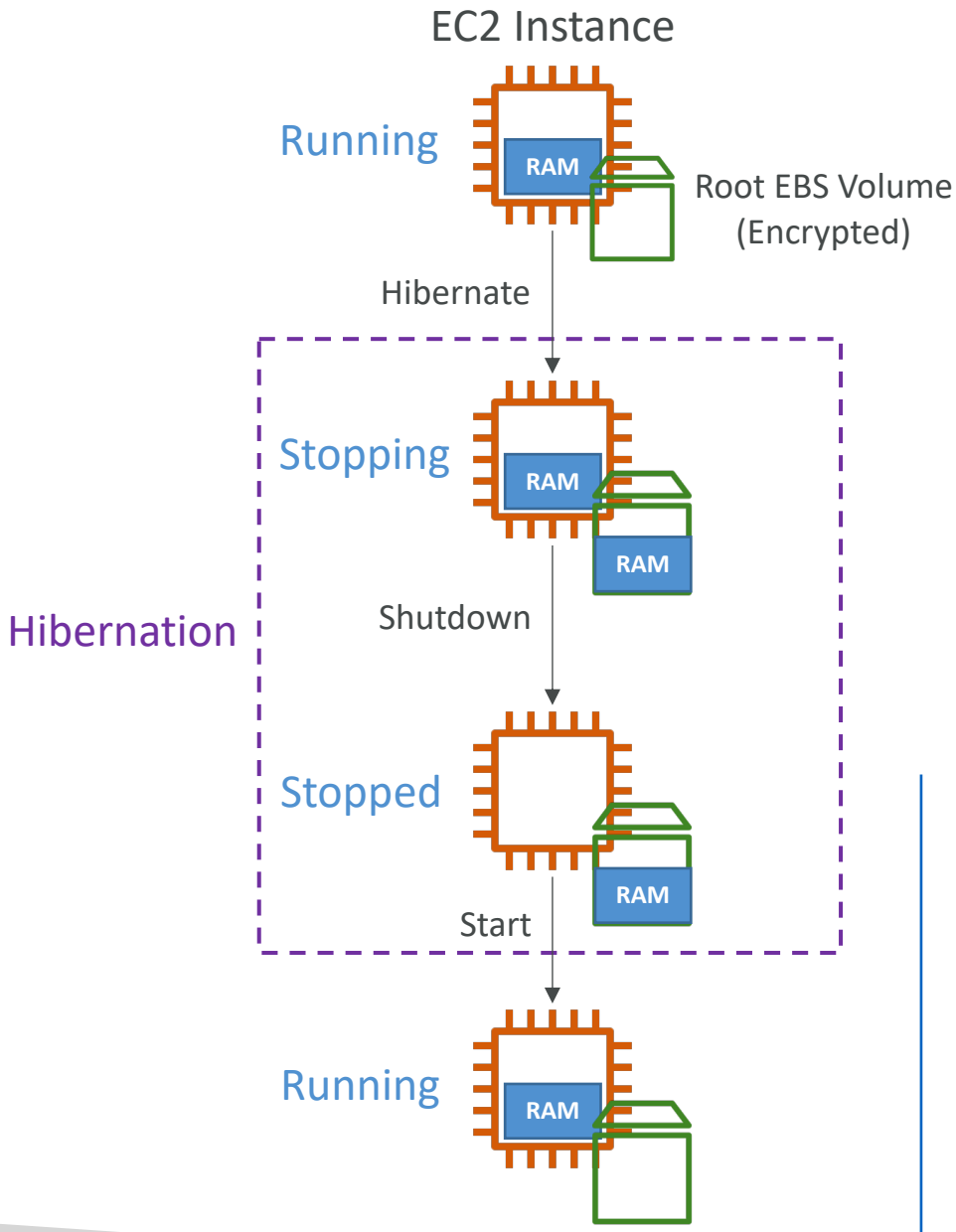
- We know we can stop, terminate instances
  - **Stop** – the data on disk (EBS) is kept intact in the next start
  - **Terminate** – any EBS volumes (root) also set-up to be destroyed is lost

- On start, the following happens:
  - First start: the OS boots & the EC2 User Data script is run
  - Following starts: the OS boots up
  - Then your application starts, caches get warmed up, and that can take time!

EC2 Hibernate

- Introducing **EC2 Hibernate**:
  - The in-memory (RAM) state is preserved
  - The instance boot is much faster!
    (the OS is not stopped / restarted)
  - Under the hood: the RAM state is written to a file in the root EBS volume
  - The root EBS volume must be encrypted

- **Use cases:**
  - Long-running processing
  - Saving the RAM state
  - Services that take time to initialize



EC2 Hibernate – Good to know

- **Supported Instance Families –** C3, C4, C5, I3, M3, M4, R3, R4, T2, T3, …
- **Instance RAM Size** – must be less than 150 GB.
- **Instance Size** – not supported for bare metal instances.
- **AMI –** Amazon Linux 2, Linux AMI, Ubuntu, RHEL, CentOS & Windows…
- **Root Volume –** must be EBS, encrypted, not instance store, and large
- Available for **On-Demand**, **Reserved** and **Spot** Instances

- An instance can **NOT** be hibernated more than 60 days